

Cascade Selection on the H_4 Branch: An Auxiliary Block Graph Laplacian and a Witnessed (O1) Discharge on V_{600}

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Abstract

We give a finite-level witnessed (O1) tuple for the cascade-mechanism consumer-API obligation [4, §3.4] on the H_4 branch at every finite top-rung level $N \geq 0$, parametrised by a free choice of $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$. The witness operator is a new auxiliary block graph Laplacian L_N^{block} on the finite-level vertex-sector Hilbert space X_N^{0,H_4} of [1, Definition **def:Xn-0**], introduced in this paper and explicitly distinct from [2]’s vertex/top-left block $A_n^{\text{vertex}} = d_n^* d_n$ (which acts trivially off the $\text{Im}(\mathbb{O})$ direction in F^0) and from [3]’s $\tilde{A}_n$ (which lives in an abstract scalar pure-midpoint reduction).

The paper proves five ingredients: L_N^{block} is bounded self-adjoint and positive semi-definite on X_N^{0,H_4} , with an explicit Chung-style operator-norm bound (Proposition 3.1); $G_N^{H_4}$ is connected at every level and $\ker L_N^{\text{block}} \cong F^0$ via constant-function embedding (32-dimensional, Theorem 4.2); the gradient flow $e^{-tL_N^{\text{block}}}$ converges to the kernel projection with exponential rate governed by the spectral gap $\lambda_+(L_N^{\text{block}}) > 0$ (uniform over initial data at fixed N , Theorem 5.2); $\dim_{\mathbb{R}}(F^0)^{W(H_4)} = 28$ by irreducibility of the standard H_4 -reflection representation (Theorem 6.2); and the resulting tuple satisfies (O1) for the chosen ξ^* and chosen top-rung initial datum (Theorem 7.2), with the standing analytic preconditions (O0) verified at the top-rung level (Proposition 7.1).

We do *not* discharge (O2), (O3), the inverse limit, the D_4 branch, or the strong sense of selection that would pick a single ray in $(F^0)^{W(H_4)}$. The latter is named **(H-uniqueness)** and lies outside the cascade-closure consumer API as stated in [4, §3.4 lines 637–639].

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1 Recap and scope

1.1 The narrow goal

This paper is a single-target follow-up to [4]: we supply a witnessed (O1) tuple for the cascade-closure consumer API on the H_4 branch at every finite top-rung level $N \geq 0$. The [4, §3.4] consumer API consists of four *separate* obligations (O0), (O1), (O2), (O3) ([4, lines 594–627]); we discharge (O1) and verify the top-rung analytic pieces of (O0) as a standing precondition (the inter-rung pushforward bonding map needed by the rung-sequence formulation of (O0) is available from [1, Definition `def:p0`, Theorem `thm:bonding`] but is not used in the present (O1)-witness construction). (O2) and (O3) are not addressed and remain in their current [3] status (i.e., (O2) *as stated* remains false; (O3) holds in the abstract scalar pure-midpoint model only).

The (O1) discharge is *parametrised* by a free choice of $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$ (the $W(H_4)$ -fixed subspace, of explicit real dimension 28, Theorem 6.2). For each fixed ξ^* we exhibit an explicit five-tuple $(\Phi_N, R_N, \Psi_N^t, \phi_N^*, \phi_N^{(0)})$ satisfying every part of (O1) ([4, lines 609–617]). The freedom in ξ^* is the (O1)-witness parameter; the [4, §3.4 lines 637–639] statement “selection of ϕ_N^* itself is not provided by cascade closure and is the successor paper’s own task” is satisfied by our construction supplying *a* ϕ_N^* . Reducing to a single ray in $(F^0)^{W(H_4)} \setminus \{0\}$ requires the named open hypothesis (**H-uniqueness**) (§10), which lies *outside* the cascade-closure API.

1.2 Explicit non-claims

Remark 1.1 (Non-claims box). This paper does *not* establish any of the following:

- (i) **Single-state selection.** ξ^* is freely chosen from $(F^0)^{W(H_4)} \setminus \{0\}$ (a punctured 28-dimensional fixed subspace); reducing to a single ray requires (**H-uniqueness**) (§10). The cascade-closure consumer API does not provide single-ray selection.
- (ii) (O2), (O3), or the full (O0)–(O3) **API.** (O0) is verified only at the top-rung level as a standing precondition for the witnessed (O1) tuple (the full rung-sequence formulation of (O0) is not built here; the required inter-rung pushforward bonding map is recorded as available from

[1, Definition `def:p0`, Theorem `thm:bonding`] but not used). (O2) as stated remains false ([3, Proposition `prop:strict-fails`]), and (O3) holds only in the abstract scalar pure-midpoint model ([3, Theorem `thm:03-discharge`]). Neither is touched here.

- (iii) **Inverse-limit (O1).** We work strictly at finite top-rung level N ; lifting to $N \rightarrow \infty$ would require the named open hypotheses of [3, §`sec:scope-and-gap`], namely (L1) (boundary-vertex-incidence agreement of the level- $(n+1)$ vertex Laplacian with the abstract scalar model up to a midpoint-supported defect) and (L3) (energy-saturation of the F^0 -valued harmonic-extension operator H_{n+1}). Neither is invoked here.
- (iv) **D_4 branch.** The construction uses the H_4 -branch action of [1, Definition `def:coxeter`] in a load-bearing way (clause 4); a D_4 -branch analogue is a separate paper.
- (v) **Identification with upstream operators.** L_N^{block} is a new auxiliary operator introduced here. Theorems about it do not transfer to [2]’s vertex/top-left block $A_n^{\text{vertex}} = d_n^* d_n$ or [3]’s $\tilde{A}_n$.
- (vi) **$W(H_4)$ -non-equivariant initial data.** Convergence still holds (Theorem 5.2), but the limit lies in the full 32-dimensional $\ker L_N^{\text{block}}$, not in the 28-dimensional $\iota_N((F^0)^{W(H_4)})$.
- (vii) **Continuum limit / target-side correspondence theorem.** No PDE identification, no metric / spectral / Hecke / cohomology correspondence.
- (viii) **Selection mechanism in the strong sense.** “Cascade closure as selection mechanism” is overstatement. This paper provides a top-rung (O1) witness for a chosen ξ^* on the H_4 branch at finite N , no more.

1.3 Inputs and notation

We use the following inputs without re-deriving:

- [1] (paper P3): the finite-level vertex-sector Hilbert space $X_n^{0,\bullet} = \ell^2(V(G_n^\bullet); F^0)$ (`def:Xn-0`, `prop:Xn-Hilbert`); the Schläfli-style refinement step (`def:refinement`); the base graph $G_0^{H_4}$ on V_{600} (`def:base-graphs`); the sector fibres $F^0 = V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8$ of $\mathbb{R}$ -dimension 32 (`not:fibres`); the branchwise Coxeter action $\rho_{F^0}^\bullet$ (`def:coxeter`); Coxeter unitarity (`lem:coxeter-unitary`); the vertex bonding map p^0 (`def:p0`, `thm:bonding`).
- [2] (paper P4): the discrete coboundary d_n which projects F^0 to $F^1 = \text{Im}(\mathbb{O})$ via $\pi_{F^0 \rightarrow F^1}$ (`def:coboundary`); the *vertex* (top-left) block $A_n^{\text{vertex}} := d_n^* d_n$ on $X_n^{0,\bullet}$ (`def:An`); the finite-dimensional ODE / matrix-exponential argument for flow existence (`thm:flow-exists`).
- [3]: the abstract scalar pure-midpoint refinement model (`def:abstract-model`); the scaled curvature operator $\tilde{A}_n := 2^n A_n^{\text{vertex}}$ in that abstract scalar model (`def:Atilde`); the open lift hypotheses (L1), (L3) (`sec:scope-and-gap`).
- [4]: the cascade-closure consumer API (O0)–(O3) (§3.4 lines 594–627); the explicit “selection is the successor’s task” framing (§3.4 lines 637–639).
- [5] (paper P2): the V_{600} vertex set (`def:V600`); the icosian ring (`def:icosian-ring`); $V_{24} \subset V_{600}$ (`prop:24-in-600`); the explicit V_{600} graph Laplacian spectrum (`tab:P2-spectrum`), used only in the $N = 0$ worked numerics of §8.

Classical references invoked: [6] for H_4 reflection-group irreducibility; [7] for the scalar graph Laplacian on multigraphs; [8], [9] for finite-dimensional spectral / matrix-exponential arguments; [10] for the 600-cell edge graph.

2 The auxiliary block graph Laplacian L_N^{block}

2.1 Why we need a new operator

The cascade-closure consumer API of [4, §3.4] requires the supplied R_N and flow Ψ_N^t to be the successor paper's own choice; in particular, R_N is not required to be P4's closure functional $F_n = \alpha R_n + \beta E_n - \gamma Q_n$ ([2, Definition `def:Fn`]). We exploit this freedom: P4's vertex/top-left block $A_n^{\text{vertex}} := d_n^* d_n$ ([2, Definition `def:An`]) acts non-trivially only on the $\text{Im}(\mathbb{O})$ direction within the V_8 fibre summand, because P4's coboundary $d_n: X_n^0 \rightarrow X_n^1$ first projects F^0 onto $F^1 = \text{Im}(\mathbb{O})$ via the canonical projection $\pi_{F^0 \rightarrow F^1}$ ([2, Definition `def:coboundary`]). Hence

$$\dim_{\mathbb{R}} \ker A_n^{\text{vertex}} = 25 |V(G_n^\bullet)| + 7$$

on a connected graph $G_n^\bullet$ (the 25 counts $\dim V_{H_4} + \dim V_{D_4} + \dim V_{16} = 4 + 4 + 16$ plus the real-octonion direction $\dim \mathbb{R} \cdot 1 = 1$, all of which d_n ignores; the additional +7 counts constant $\text{Im}(\mathbb{O})$ -functions). That kernel is too large for a clean (O1) witness.

We need an operator whose kernel is exactly the F^0 -valued constants on a connected refined graph (dimension $\dim_{\mathbb{R}} F^0 = 32$); the simple block-diagonal extension of the scalar graph Laplacian to F^0 -valued functions does the job and is what we introduce below.

2.2 Definition

Definition 2.1 (Labelled-multiset scalar graph Laplacian). Let $G_N^{H_4} = (V(G_N^{H_4}), E(G_N^{H_4}))$ be the level- N Schläfli-style refined graph of [1, Definition `def:refinement`], with base $G_0^{H_4}$ on V_{600} ([1, Definition `def:base-graphs`]). By [1, Definition `def:refinement`], $E(G_N^{H_4})$ is a *labelled-edge multiset*: each generated edge carries a class tag $\in \{(a), (b), (c)\}$ and a parent-face tag where applicable, so two edges with the same unordered endpoint pair but different tags are distinct elements of $E(G_N^{H_4})$.

The *labelled-multiset combinatorial scalar graph Laplacian* L_N^{cl} on $V(G_N^{H_4})$, acting on $\mathbb{R}$ -valued functions, is

$$(L_N^{\text{cl}} f)(v) := \sum_{e \in E(G_N^{H_4}), v \in e} (f(v) - f(\text{op}_e(v))),$$

where for each $e = \{v, \text{op}_e(v)\}$ in the multiset, $\text{op}_e(v)$ denotes the other endpoint of the labelled edge e . Each labelled edge $e \in E(G_N^{H_4})$ contributes once with multiplicity 1; parallel labelled edges with coinciding endpoints are summed separately.

Equivalently, $L_N^{\text{cl}} = D_N - W_N$ where the diagonal $D_N(v) := |\{e \in E(G_N^{H_4}) : v \in e\}| = \deg_{G_N^{H_4}}(v)$ uses the labelled-multiset degree count, and $W_N(v, u) := |\{e \in E(G_N^{H_4}) : e = \{v, u\}\}|$ is the number of labelled edges joining v to u (an integer ≥ 0). This is the standard Chung convention ([7, Ch. 1]) extended to multigraphs.

For $N = 0$ all base edges of $G_0^{H_4}$ on V_{600} have multiplicity 1 (the 600-cell edge graph is simple, [10, §22.7]), so $W_0(v, u) \in \{0, 1\}$ and L_0^{cl} reduces to the standard simple graph Laplacian on the 600-cell edge graph.

Definition 2.2 (Block graph Laplacian). The *block graph Laplacian on the H_4 branch at level N* is the bounded $\mathbb{R}$ -linear operator L_N^{block} on the finite-dimensional Hilbert space $X_N^{0,H_4} = \ell^2(V(G_N^{H_4}); F^0)$ ([1, Definition `def:Xn-0`]) defined fibrewise by

$$L_N^{\text{block}}(\phi)(v) := (L_N^{\text{cl}}\phi^{(H_4)})(v) \oplus (L_N^{\text{cl}}\phi^{(D_4)})(v) \oplus (L_N^{\text{cl}}\phi^{(16)})(v) \oplus (L_N^{\text{cl}}\phi^{(8)})(v),$$

where $\phi(v) = \phi^{(H_4)}(v) \oplus \phi^{(D_4)}(v) \oplus \phi^{(16)}(v) \oplus \phi^{(8)}(v)$ in the orthogonal decomposition $F^0 = V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8$ of [1, Notation `not:fibres`], and L_N^{cl} acts componentwise on each summand independently via Definition 2.1.

2.3 Relation to upstream operators

Remark 2.3 (L_N^{block} is not an upstream operator). We do *not* claim that L_N^{block} equals or is a restriction of any operator from [2] or [3]. Specifically:

- The top-left block $d_n^* d_n$ of P4's A_n , denoted here by A_n^{vertex} (a local shorthand of the present paper, not P4's literal symbol; P4 itself names A_n for the full block operator, [2, Definition `def:An`]), factors through the $\text{Im}(\mathbb{O})$ projection of P4's coboundary ([2, Definition `def:coboundary`]): informally, $A_n^{\text{vertex}} = \iota_{F^1 \rightarrow F^0} \circ L_N^{\text{cl}} \circ \pi_{F^0 \rightarrow F^1}$. Compared with L_N^{block} , this replaces three of the four fibre blocks (V_{H_4} , V_{D_4} , V_{16} , plus the real-octonion direction in V_8) with zero. Hence L_N^{block} is a strictly larger operator on X_N^{0,H_4} .
- [3]'s scaled curvature operator $\tilde{A}_n := 2^n A_n^{\text{vertex}}$ ([3, Definition `def:Atilde`]) is the *scaled scalar abstract-model analogue* of P4's vertex Laplacian: [3] works in the abstract scalar pure-midpoint reduction ([3, Definition `def:abstract-model`]), where the underlying state space is scalar-valued (not F^0 -valued) on a pure-midpoint refinement of V_{600} . The full-fibre L_N^{block} on X_N^{0,H_4} used here is in a different scope from both [3] (which has no fibre F^0) and [2] (which restricts to the $\text{Im}(\mathbb{O})$ direction).

L_N^{block} is introduced in this paper for the (O1) witness. The simple block-diagonal construction is one natural choice; we do not claim minimality.

3 Self-adjointness, positive semi-definiteness, and bound

Proposition 3.1 (Clause (1): operator structure of L_N^{block}). *For every $N \geq 0$, L_N^{block} is bounded, self-adjoint, and positive semi-definite on X_N^{0,H_4} , with operator-norm bound*

$$\|L_N^{\text{block}}\|_{\text{op}} \leq 2 \Delta(G_N^{H_4})$$

where $\Delta(G_N^{H_4}) := \max_{v \in V(G_N^{H_4})} \deg_{G_N^{H_4}}(v)$ is the maximum labelled-multiset degree.

Proof. Self-adjointness and positive semi-definiteness of the scalar Laplacian L_N^{cl} follow from the standard identity

$$\langle L_N^{\text{cl}} f, g \rangle_{\ell^2(V(G_N^{H_4}); \mathbb{R})} = \sum_{e \in E(G_N^{H_4})} (f(h(e)) - f(t(e)))(g(h(e)) - g(t(e)))$$

(for any orientation $e \mapsto (t(e), h(e))$ of each labelled edge; the sum is symmetric in the unoriented edge), which gives $\langle L_N^{\text{cl}} f, g \rangle = \langle f, L_N^{\text{cl}} g \rangle$ and $\langle L_N^{\text{cl}} f, f \rangle = \sum_e (f(h(e)) - f(t(e)))^2 \geq 0$ ([7, Ch. 1] for the multigraph extension).

The block-diagonal construction $L_N^{\text{block}} = L_N^{\text{cl}} \otimes \text{id}_{F^0}$ (after identifying $X_N^{0,H_4} = \ell^2(V(G_N^{H_4}); \mathbb{R}) \otimes_{\mathbb{R}} F^0$ orthogonally) preserves both self-adjointness and positive semi-definiteness.

For the bound, the standard scalar estimate $\|L_N^{\text{cl}}\|_{\text{op}} \leq 2\Delta(G_N^{H_4})$ ([7, Ch. 1], multigraph case) extends to the block operator with the same constant via the tensor-product operator-norm identity: under the orthogonal identification $X_N^{0,H_4} \cong \ell^2(V(G_N^{H_4}); \mathbb{R}) \otimes_{\mathbb{R}} F^0$ of Definition 2.2, $L_N^{\text{block}} = L_N^{\text{cl}} \otimes \text{id}_{F^0}$, so $\|L_N^{\text{block}}\|_{\text{op}} = \|L_N^{\text{cl}}\|_{\text{op}} \cdot \|\text{id}_{F^0}\|_{\text{op}} = \|L_N^{\text{cl}}\|_{\text{op}}$ ([9, Ch. 4.2]). $\square$

4 Connectivity and the kernel theorem

Lemma 4.1 (Connectivity of the refined H_4 graph). *For every $N \geq 0$, the (labelled-multiset) graph $G_N^{H_4}$ is connected.*

Proof. Induction on N .

Base case $N = 0$. $G_0^{H_4}$ has vertex set V_{600} and edges joining 600-cell vertices at the standard 600-cell edge distance ([1, Definition `def:base-graphs`], [10, §22.7]). The 600-cell edge graph is connected; this is classical and equivalent to $W(H_4)$ -orbit transitivity on V_{600} via edge-adjacent unit icosians.

Inductive step. Assume $G_n^{H_4}$ is connected. By [1, Definition `def:refinement`], $V(G_{n+1}^{H_4}) = V(G_n^{H_4}) \sqcup M_n^{H_4} \sqcup C_n^{H_4}$ (vertices, midpoints, centroids), and the edge set $E(G_{n+1}^{H_4})$ consists of three classes:

- Class (a): for each $e = \{u, v\} \in E_n^{H_4}$, two subdividing edges $\{u, m_e\}$ and $\{m_e, v\}$ (using the midpoint m_e).
- Class (b): for each face $f \in F_n^{H_4}$ with edge sequence $(e_1, \dots, e_k)$, midpoint-centroid edges $\{m_{e_i}, c_f\}$.
- Class (c): midpoint-midpoint diagonal edges $\{m_{e_i}, m_{e_{i+1}}\}$ closing each level- $(n+1)$ triangle face.

Old vertices first: every $G_n^{H_4}$ -edge $e = \{u, v\}$ is replaced in $G_{n+1}^{H_4}$ by the length-two path $u - m_e - v$ via two Class-(a) edges. Hence any $G_n^{H_4}$ -path $u = v_0, v_1, \dots, v_r = w$ joining old vertices $u, w \in V(G_n^{H_4})$ is replaced by a length- $2r$ path $v_0, m_{e_1}, v_1, m_{e_2}, \dots, m_{e_r}, v_r$ in $G_{n+1}^{H_4}$, so $V(G_n^{H_4}) \subset V(G_{n+1}^{H_4})$ remains mutually connected by the inductive hypothesis on $G_n^{H_4}$.

New vertices: each level- $(n+1)$ midpoint m_e is adjacent to both endpoints of its parent edge $e \in E_n^{H_4}$ via Class-(a) edges, hence to $V(G_n^{H_4})$. Each level- $(n+1)$ centroid c_f is adjacent to every midpoint m_{e_i} of every edge in $\partial_n^\bullet(f)$ via Class-(b) edges, hence (through the already-attached midpoints) to $V(G_n^{H_4})$. Therefore every vertex of $V(G_{n+1}^{H_4})$ is in the same connected component as the (mutually connected) old vertex set, which gives connectivity of $G_{n+1}^{H_4}$. Class-(c) edges are not needed for the inductive step. $\square$

Theorem 4.2 (Clause (2): kernel structure of L_N^{block}). *For every $N \geq 0$, the constant-function embedding*

$$\iota_N: F^0 \rightarrow \ker L_N^{\text{block}}, \quad \iota_N(\xi)(v) := \xi \quad \text{for all } v \in V(G_N^{H_4}),$$

is an $\mathbb{R}$ -linear isomorphism. In particular, $\dim_{\mathbb{R}} \ker L_N^{\text{block}} = \dim_{\mathbb{R}} F^0 = 32$.

Proof. The kernel of the scalar Laplacian on a connected (multi)graph is one-dimensional, spanned by the constant function ([7, Ch. 1], multigraph case). By Lemma 4.1, $G_N^{H_4}$ is connected, so $\ker L_N^{\text{cl}} = \mathbb{R} \cdot \mathbf{1}_{V(G_N^{H_4})}$.

By the tensor identification of Definition 2.2, $L_N^{\text{block}} = L_N^{\text{cl}} \otimes \text{id}_{F^0}$ on $X_N^{0,H_4} \cong \ell^2(V(G_N^{H_4}); \mathbb{R}) \otimes_{\mathbb{R}} F^0$, so

$$\ker L_N^{\text{block}} = (\ker L_N^{\text{cl}}) \otimes_{\mathbb{R}} F^0 = (\mathbb{R} \cdot \mathbf{1}_{V(G_N^{H_4})}) \otimes_{\mathbb{R}} F^0 \cong F^0,$$

where the last isomorphism sends $\mathbf{1} \otimes \xi$ to $\xi \in F^0$. This isomorphism is exactly ι_N (constant-function embedding $\xi \mapsto (v \mapsto \xi)$).

$\mathbb{R}$ -linearity of ι_N is by construction; injectivity is immediate; surjectivity onto $\ker L_N^{\text{block}}$ is the above identification. $\square$

5 Convergence theorem

Notation 5.1 (Spectral gap). For a positive semi-definite self-adjoint operator T on a finite-dimensional Hilbert space whose positive spectrum is nonempty (i.e., $T \neq 0$), write

$$\lambda_+(T) := \min\{\lambda \in \text{spec}(T) : \lambda > 0\}$$

for the smallest positive eigenvalue (the spectral gap). When $\dim \ker T > 1$, $\lambda_+(T)$ is strictly larger than the “second eigenvalue” $\lambda_2(T) = 0$, and we use λ_+ throughout to avoid confusion.

Theorem 5.2 (Clause (3): convergence of the gradient flow). *For every $N \geq 0$ and every $\Phi(0) \in X_N^{0,H_4}$, the operator semigroup $e^{-tL_N^{\text{block}}}$ satisfies*

$$\lim_{t \rightarrow \infty} e^{-tL_N^{\text{block}}} \Phi(0) = P_{\ker L_N^{\text{block}}} \Phi(0) \quad \text{in } X_N^{0,H_4}\text{-norm,}$$

with exponential bound, uniform over initial data $\Phi(0)$ at fixed N (no claim about uniformity in N),

$$\|e^{-tL_N^{\text{block}}} \Phi(0) - P_{\ker L_N^{\text{block}}} \Phi(0)\|_{X_N^{0,H_4}} \leq e^{-t\lambda_+(L_N^{\text{block}})} \|\Phi(0) - P_{\ker L_N^{\text{block}}} \Phi(0)\|_{X_N^{0,H_4}}$$

for every $t \geq 0$, where $\lambda_+(L_N^{\text{block}}) = \lambda_+(L_N^{\text{cl}}) > 0$ (positivity is by Lemma 4.1: the connected, nontrivial multigraph $G_N^{H_4}$ has a strictly positive Fiedler value; the block extension repeats each scalar eigenvalue with multiplicity 32, leaving the smallest positive eigenvalue unchanged).

Proof. By Proposition 3.1, L_N^{block} is bounded self-adjoint and positive semi-definite on the finite-dimensional Hilbert space X_N^{0,H_4} . By the spectral theorem ([8, Ch. VIII.5], [9, Ch. 6.2]), there is an orthogonal decomposition

$$X_N^{0,H_4} = \bigoplus_{\lambda \in \text{spec}(L_N^{\text{block}})} E_{\lambda}(L_N^{\text{block}}),$$

where $E_0 = \ker L_N^{\text{block}}$ and every other E_{λ} has $\lambda \geq \lambda_+(L_N^{\text{block}})$. For $\Phi(0) = \sum_{\lambda} \Phi_{\lambda}(0)$ with $\Phi_{\lambda}(0) \in E_{\lambda}$,

$$e^{-tL_N^{\text{block}}} \Phi(0) = \sum_{\lambda} e^{-t\lambda} \Phi_{\lambda}(0) = \Phi_0(0) + \sum_{\lambda > 0} e^{-t\lambda} \Phi_{\lambda}(0).$$

The first term is $P_{\ker} \Phi(0)$, constant in t . The second term is bounded by

$$\left\| \sum_{\lambda > 0} e^{-t\lambda} \Phi_{\lambda}(0) \right\|^2 = \sum_{\lambda > 0} e^{-2t\lambda} \|\Phi_{\lambda}(0)\|^2 \leq e^{-2t\lambda_+} \sum_{\lambda > 0} \|\Phi_{\lambda}(0)\|^2 = e^{-2t\lambda_+} \|\Phi(0) - P_{\ker} \Phi(0)\|^2,$$

which gives the stated exponential bound on taking square roots.

The eigenvalue identity $\lambda_+(L_N^{\text{block}}) = \lambda_+(L_N^{\text{cl}})$ follows from $\text{spec}(L_N^{\text{block}}) = \text{spec}(L_N^{\text{cl}})$ (set-equality; multiplicities multiply by $\dim F^0 = 32$ but the set of distinct eigenvalues is unchanged), which is a standard property of block-diagonal operators ([9, Ch. 6.4]). $\square$

6 The $W(H_4)$ -fixed subspace count

Lemma 6.1 (Standard H_4 -reflection representation has trivial fixed subspace). *The standard reflection representation of the finite irreducible Coxeter group $W(H_4)$ on $V_{H_4} \cong \mathbb{H} \cong \mathbb{R}^4$ satisfies $V_{H_4}^{W(H_4)} = \{0\}$.*

Proof. $W(H_4)$ is a finite irreducible Coxeter group; the standard reflection representation of any finite irreducible Coxeter group is irreducible ([6, Ch. 5]; see also [6, Ch. 2] for H_4 specifically). The reflection representation of a finite irreducible Coxeter group is faithful and is not the trivial representation ([6, Ch. 5]). The fixed subspace $V_{H_4}^{W(H_4)} \subset V_{H_4}$ is a $W(H_4)$ -invariant subspace; by irreducibility it is either $\{0\}$ or all of V_{H_4} . The latter would force every reflection to act as the identity, contradicting faithfulness (or non-triviality). Hence $V_{H_4}^{W(H_4)} = \{0\}$. $\square$

Theorem 6.2 (Clause (4): the $W(H_4)$ -fixed subspace of F^0). *Under the branchwise Coxeter representation $\rho_{F^0}^{H_4}$ of [1, Definition `def:coxeter`] (standard H_4 -reflection representation on V_{H_4} , trivial on V_{D_4}, V_{16}, V_8),*

$$(F^0)^{W(H_4)} = V_{H_4}^{W(H_4)} \oplus V_{D_4} \oplus V_{16} \oplus V_8 = \{0\} \oplus V_{D_4} \oplus V_{16} \oplus V_8,$$

so $\dim_{\mathbb{R}}(F^0)^{W(H_4)} = 0 + 4 + 16 + 8 = 28$.

Proof. The branchwise representation $\rho_{F^0}^{H_4} = \rho_{V_{H_4}}^{H_4} \oplus \rho_{V_{D_4}}^{H_4} \oplus \rho_{V_{16}}^{H_4} \oplus \rho_{V_8}^{H_4}$ acts on each summand independently ([1, Definition `def:coxeter`] fixes the standard reflection representation on V_{H_4} and the trivial representation on the other three summands). Hence

$$(F^0)^{W(H_4)} = V_{H_4}^{W(H_4)} \oplus V_{D_4}^{W(H_4)} \oplus V_{16}^{W(H_4)} \oplus V_8^{W(H_4)}.$$

By Lemma 6.1, $V_{H_4}^{W(H_4)} = \{0\}$. By the trivial representation on the other summands, $V_{D_4}^{W(H_4)} = V_{D_4}$ (dim 4), $V_{16}^{W(H_4)} = V_{16}$ (dim 16), $V_8^{W(H_4)} = V_8$ (dim 8). Summing dimensions, $\dim_{\mathbb{R}}(F^0)^{W(H_4)} = 0 + 4 + 16 + 8 = 28$. $\square$

7 (O0) verification and (O1) witnessed discharge

7.1 Top-rung analytic preconditions and recorded inter-rung bonding map

Proposition 7.1 (Top-rung analytic preconditions and recorded inter-rung bonding map). *The top-rung analytic components of the cascade-mechanism standing preconditions (O0) of [4, §3.4 lines 595–608] hold for the five-tuple supplied in Theorem 7.2 below; the inter-rung bonding map needed by the rung-sequence formulation of (O0) is recorded as available from [1, Definition `def:p0`, Theorem `thm:bonding`] but is not used here. Specifically:*

- (O0.1) Finite-dimensional Hilbert structure of $\Phi_N = X_N^{0,H_4}$. *By [1, Proposition `prop:Xn-Hilbert`], X_N^{0,H_4} is a finite-dimensional real Hilbert space of dimension $|V(G_N^{H_4})| \cdot 32$.*
- (O0.2) $R_N \in C^1(\Phi_N)$. *The quadratic form $R_N(\phi) = \frac{1}{2}\langle \phi, L_N^{\text{block}} \phi \rangle$ on a finite-dimensional Hilbert space is real-analytic (in particular C^1); its gradient is $\nabla R_N(\phi) = L_N^{\text{block}} \phi$ by Proposition 3.1 (self-adjointness).*

- (O0.3) Well-posed gradient flow $\Psi_N^t = e^{-tL_N^{\text{block}}}$. By the same finite-dimensional ODE / matrix-exponential argument as [2, Theorem `thm:flow-exists`], applied here to the bounded self-adjoint L_N^{block} on the finite-dimensional Hilbert space X_N^{0,H_4} , the initial-value problem $\dot{\Phi}_N(t) = -L_N^{\text{block}}\Phi_N(t)$, $\Phi_N(0) = \Phi(0)$ has a unique global solution $\Phi_N(t) = e^{-tL_N^{\text{block}}}\Phi(0)$.
- (O0.4) C^1 pushforward bonding maps (recorded only). The (O1) witness below uses only top-rung residual-zero and convergence data; no inter-rung pushforward is invoked. For completeness, when the cascade-mechanism API invokes an inter-rung pushforward $\pi_{k+1,k}^\sharp: \Phi_{k+1} \rightarrow \Phi_k$ for downstream propagation ([4, §3.4 lines 600–605]), [1, Definition `def:p0`, Theorem `thm:bonding`] supplies a bounded $\mathbb{R}$ -linear (hence C^1 , in fact C^ω) vertex restriction map $p_{k+1,k}^0: X_{k+1}^{0,H_4} \rightarrow X_k^{0,H_4}$ between adjacent levels. This paper does not use it for the (O1) discharge but records its availability for downstream consumers wishing to instantiate the full rung-sequence formulation of (O0).

Proof. Each item follows directly from the cited result, as stated. $\square$

7.2 (O1) witnessed discharge: the named theorem

Theorem 7.2 (Witnessed (O1) discharge on the H_4 branch at finite top-rung level N , parametrised by ξ^*). Let $N \geq 0$ and let L_N^{block} be the block graph Laplacian of Definition 2.2. For each fixed

$$\xi^* \in (F^0)^{W(H_4)} \setminus \{0\},$$

where $(F^0)^{W(H_4)}$ is the $W(H_4)$ -fixed subspace of F^0 of explicit real dimension 28 (Theorem 6.2), set:

- $\Phi_N := X_N^{0,H_4}$ ([1, Definition `def:Xn-0`]);
- $R_N(\phi) := \frac{1}{2} \langle \phi, L_N^{\text{block}} \phi \rangle_{X_N^{0,H_4}}$, the quadratic-form residual associated to L_N^{block} ;
- $\Psi_N^t := e^{-tL_N^{\text{block}}}$, the gradient flow of R_N ;
- $\phi_N^* := \iota_N(\xi^*)$, the constant ξ^* -valued function on $V(G_N^{H_4})$ (Theorem 4.2);
- $\phi_N^{(0)} := \phi_N^*$.

Then the five-tuple $(\Phi_N, R_N, \Psi_N^t, \phi_N^*, \phi_N^{(0)})$ satisfies the cascade-mechanism (O1) obligation of [4, §3.4 lines 609–617] in the witnessed fixed-point sense, for the chosen ξ^* and the specified initial datum $\phi_N^{(0)} = \phi_N^*$ (no claim is made about basin-of-attraction selection from arbitrary initial data, beyond the conditional clause (O1.2) below):

- (O1.1) Top-rung residual-zero state. $\phi_N^* \in \Phi_N$ satisfies $R_N(\phi_N^*) = 0$.
- (O1.2) Convergence from a chosen initial datum. For $\phi_N^{(0)} = \phi_N^*$, $\Psi_N^t \phi_N^{(0)} = \phi_N^*$ for every $t \geq 0$ (a fixed point of the flow). More generally, for any $\tilde{\phi}_N^{(0)} \in \Phi_N$ with $P_{\ker L_N^{\text{block}}} \tilde{\phi}_N^{(0)} = \phi_N^*$, $\lim_{t \rightarrow \infty} \Psi_N^t \tilde{\phi}_N^{(0)} = \phi_N^*$ in X_N^{0,H_4} -norm with uniform exponential bound governed by $\lambda_+(L_N^{\text{block}})$ (Theorem 5.2).

The relevant top-rung analytic preconditions from (O0) are verified in Proposition 7.1; the rung-sequence formulation of (O0) is not built here (the required inter-rung bonding map is recorded as available but unused).

Proof. (O1.1). By Theorem 4.2, $\iota_N(\xi^*) \in \ker L_N^{\text{block}}$, so $L_N^{\text{block}}\phi_N^* = 0$, hence $R_N(\phi_N^*) = \frac{1}{2}\langle \phi_N^*, L_N^{\text{block}}\phi_N^* \rangle = 0$.

(O1.2). For $\phi_N^{(0)} = \phi_N^*$, $\Psi_N^t \phi_N^* = e^{-tL_N^{\text{block}}} \phi_N^* = \phi_N^*$ for all $t \geq 0$ because $\phi_N^* \in \ker L_N^{\text{block}}$ implies $L_N^{\text{block}}\phi_N^* = 0$, hence $e^{-tL_N^{\text{block}}} \phi_N^* = \sum_{k \geq 0} \frac{(-t)^k}{k!} (L_N^{\text{block}})^k \phi_N^* = \phi_N^*$ (only the $k = 0$ term survives).

For the more general statement: any $\tilde{\phi}_N^{(0)}$ with $P_{\ker L_N^{\text{block}}} \tilde{\phi}_N^{(0)} = \phi_N^*$ decomposes orthogonally as $\tilde{\phi}_N^{(0)} = \phi_N^* + \tilde{\phi}^\perp$ with $\tilde{\phi}^\perp \in (\ker L_N^{\text{block}})^\perp$. By Theorem 5.2, $\Psi_N^t \tilde{\phi}^\perp \rightarrow 0$ in norm with rate λ_+ , while $\Psi_N^t \phi_N^* = \phi_N^*$ as above. Hence $\Psi_N^t \tilde{\phi}_N^{(0)} \rightarrow \phi_N^*$ in norm. $\square$

Remark 7.3 (On the ξ^* parameter). For each $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$, Theorem 7.2 produces a witnessed (O1)-tuple. The freedom in ξ^* is the (O1)-witness parameter; the [4, §3.4 lines 637–639] statement “selection of ϕ_N^* itself is not provided by cascade closure and is the successor paper’s own task” is satisfied by our supplying $a \phi_N^*$, not the unique one. Reducing the 28-dimensional ξ^* -freedom to a single ray requires the named open hypothesis (**H-uniqueness**) (§10).

The chosen ξ^* is $W(H_4)$ -equivariant by construction ($\xi^* \in (F^0)^{W(H_4)}$), so $\iota_N(\xi^*)$ is a $W(H_4)$ -equivariant function on $V(G_N^{H_4})$. $W(H_4)$ -non-equivariant initial data $\tilde{\phi}_N^{(0)} \in X_N^{0,H_4}$ still converge by Theorem 5.2, but to the 32-dimensional full kernel projection $P_{\ker L_N^{\text{block}}} \tilde{\phi}_N^{(0)}$, which need not coincide with any constant $\iota_N(\xi^*)$ for ξ^* in the 28-dimensional $W(H_4)$ -fixed subspace.

8 Worked numerics for $N = 0$

We exhibit the explicit numerical content of Theorem 7.2 at the base level $N = 0$, where the graph $G_0^{H_4}$ on V_{600} has a fully understood spectrum imported from [5, Table `tab:P2-spectrum`].

- Vertex count: $|V(G_0^{H_4})| = |V_{600}| = 120$ ([5, Definition `def:V600`]).
- Edge count: $|E(G_0^{H_4})| = 720$ (the 600-cell edge graph has 120 vertices each of degree 12, giving $\frac{120 \cdot 12}{2} = 720$ edges; [10, §22.7]).
- Maximum labelled-multiset degree (here equal to ordinary degree since the base graph is simple): $\Delta(G_0^{H_4}) = 12$ (uniform).
- Hilbert-space dimension: $\dim_{\mathbb{R}} X_0^{0,H_4} = 120 \cdot 32 = 3840$ ([1, Proposition `prop:Xn-Hilbert`]).
- Block-Laplacian operator-norm bound: $\|L_0^{\text{block}}\|_{\text{op}} \leq 2 \cdot 12 = 24$ (Proposition 3.1). The actual operator norm is $\|L_0^{\text{block}}\|_{\text{op}} = 9 + 3\sqrt{5} \approx 15.708$ (largest eigenvalue from [5, Table `tab:P2-spectrum`]).
- Kernel dimension: $\dim_{\mathbb{R}} \ker L_0^{\text{block}} = 32$ (Theorem 4.2; the trivial irrep multiplicity is 1 for the scalar Laplacian, so $1 \cdot 32 = 32$ in the block extension).
- Spectral gap: $\lambda_+(L_0^{\text{block}}) = \lambda_+(L_0^{\text{cl}}) = 9 - 3\sqrt{5} \approx 2.2918$ (smallest positive eigenvalue from [5, Table `tab:P2-spectrum`], the **2** irrep entry).
- $W(H_4)$ -fixed subspace of F^0 : $\dim_{\mathbb{R}}(F^0)^{W(H_4)} = 28$ (Theorem 6.2).
- ξ^* -parametrised constant subspace inside $\ker L_0^{\text{block}}$: $\dim_{\mathbb{R}} \iota_0((F^0)^{W(H_4)}) = 28$ (image of injective ι_0).

The convergence rate constant is therefore

$$\lambda_+(L_0^{\text{block}}) = 9 - 3\sqrt{5},$$

giving the explicit decay bound $\|\Phi(t) - P_{\ker}\Phi(0)\| \leq e^{-(9-3\sqrt{5})t} \|\Phi(0) - P_{\ker}\Phi(0)\|$ for any initial datum $\Phi(0) \in X_0^{0,H_4}$.

Remark 8.1 (Numerical verification (optional)). A one-line numerical check of the spectrum is straightforward: build the 120×120 scalar Laplacian $L_0^{\text{cl}} = D_0 - W_0$ from the 600-cell adjacency matrix and compute its eigenvalues. The result reproduces [5, Table `tab:P2-spectrum`]: nine distinct eigenvalues $\{0, 9 - 3\sqrt{5}, 10 - 2\sqrt{5}, 9, 12, 14, 10 + 2\sqrt{5}, 15, 9 + 3\sqrt{5}\}$ with multiplicities $\{1, 4, 9, 16, 25, 36, 9, 16, 4\}$ summing to 120. The block-Laplacian spectrum is the same set of eigenvalues, each with multiplicity multiplied by 32.

9 What this paper does not do

For convenience we collect the explicit non-claims of Remark 1.1 as bullet points, with cross-references to where in this paper or in upstream sources each non-claim is located:

- **Single-state selection.** See §10 (**H-uniqueness**) below; reducing the 28-dimensional ξ^* freedom to a single ray is the next-paper task.
- **(O2) and (O3).** Inherited from [3]: (O2) as stated remains false ([3, Proposition `prop:strict-fails`]), and (O3) holds only in the abstract scalar pure-midpoint model ([3, Theorem `thm:03-discharge`]).
- **Inverse-limit (O1).** Lifting Theorem 7.2 to $N \rightarrow \infty$ requires the open hypotheses (L1), (L3) of [3, §sec:scope-and-gap].
- **D_4 branch.** Theorem 6.2 and the surrounding construction use the H_4 -branch action of [1, Definition `def:coxeter`]; the D_4 analogue would have a different fixed-subspace count and is a separate paper.
- **Identification of L_N^{block} with upstream operators.** See Remark 2.3.
- **$W(H_4)$ -non-equivariant initial data convergent to $(F^0)^{W(H_4)}$.** See Remark 7.3.
- **Continuum limit / target-side correspondence.** No PDE identification; no metric, spectral, Hecke, or cohomology correspondence theorem.
- **Selection mechanism in the strong sense.** “Cascade closure as selection mechanism” is overstatement. This paper provides only a top-rung (O1) witness for a chosen ξ^* on the H_4 branch at finite N .

10 Named open hypotheses

Remark 10.1 ((**H-uniqueness**): single-ray selection). Selection of a single ray in $(F^0)^{W(H_4)} \setminus \{0\}$ requires structure beyond the cascade-closure flow. Candidates for an additional selection principle include:

- closure-energy normalisation on the kernel of L_N^{block} ;

- an additional symmetry beyond $W(H_4)$ (e.g., time-reversal, or the σ Galois action on rationality data of [1, Definition `def:sigma`]);
- a top-rung initial condition tied to a physical observable;
- a thermodynamic / variational principle on the family of admissible ξ^* .

(H-uniqueness) is named here, not discharged. Per [4, §3.4 lines 637–639], “selection of ϕ_N^* itself is not provided by cascade closure and is the successor paper’s own task” — this paper supplies *a* ϕ_N^* (parametrised by ξ^*), not the unique one. Reducing the 28-dimensional freedom to a single ray lies *outside* the cascade-closure consumer API.

Remark 10.2 ((H-physical-interpretation)). Whether $\iota_N((F^0)^{W(H_4)}) \subset X_N^{0,H_4}$ is the *physically correct* selected subspace — as opposed to a mathematical artefact of restricting to the H_4 branch with the constant-function embedding — is a physical-modelling question outside this paper’s scope. The mathematical theorem makes no claim of physical correctness or uniqueness of interpretation.

11 Citation contract and downstream handoff

11.1 What downstream papers may cite

1. Definition 2.2 (L_N^{block} as an auxiliary operator on X_N^{0,H_4} , distinct from upstream operators by Remark 2.3).
2. Proposition 3.1 (PSD self-adjointness of L_N^{block} with explicit norm bound).
3. Theorem 4.2 ($\ker L_N^{\text{block}} \cong F^0$ via ι_N).
4. Theorem 5.2 (convergence of $e^{-tL_N^{\text{block}}}$ with rate $\lambda_+(L_N^{\text{block}})$).
5. Theorem 6.2 (explicit $\dim_{\mathbb{R}}(F^0)^{W(H_4)} = 28$).
6. Theorem 7.2 (witnessed (O1) discharge for any chosen $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$).
7. Worked numerics of §8 ($N = 0$ explicit values).

11.2 What downstream papers may not cite

1. Any theorem about L_N^{block} as a theorem about P4’s vertex/top-left block $A_n^{\text{vertex}} = d_n^* d_n$ or compat’s $\tilde{A}_n$: the operators differ (Remark 2.3).
2. Single-state selection of a unique ϕ_N^* : Theorem 7.2 parametrises ϕ_N^* by $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$ (a punctured 28-dimensional fixed subspace); single-ray selection is **(H-uniqueness)** (§10).
3. Inverse-limit (O1) as $N \rightarrow \infty$: open under (L1), (L3) ([3, §sec:scope-and-gap]).
4. (O2), (O3), the full cascade-closure (O0)–(O3) API: not addressed (see Remark 1.1).
5. D_4 -branch witness: requires a separate paper.

11.3 Recommended citation phrasing

“By cascade-selection-h4 (Theorem 7.2), the cascade-mechanism consumer-API obligation ($O1$) on the H_4 branch is satisfied *in the witnessed fixed-point sense* at every finite top-rung level $N \geq 0$ by the explicit five-tuple $(X_N^{0,H_4}, \frac{1}{2}\langle \cdot, L_N^{\text{block}} \cdot \rangle, e^{-tL_N^{\text{block}}}, \iota_N(\xi^*), \iota_N(\xi^*))$ for any chosen $\xi^* \in (F^0)^{W(H_4)} \setminus \{0\}$, where L_N^{block} is the auxiliary block graph Laplacian and $(F^0)^{W(H_4)}$ has explicit real dimension 28. The construction leaves a 28-dimensional free choice of ξ^* ; single-ray selection is the named open hypothesis (**H-uniqueness**), outside the cascade-closure consumer API. No basin-of-attraction selection from arbitrary initial data is claimed.”

References

- [1] *Cascade-Classical Correspondence III: Refinement Spaces and Symmetry Actions*, paper P3. `papers/cascade-refinement-spaces/cascade-refinement-spaces.tex`. Pinned results used here, cited by label: `def:Xn-0`, `def:refinement`, `def:base-graphs`, `prop:Xn-Hilbert`, `not:fibres`, `def:coxeter`, `lem:coxeter-unitary`, `def:p0`, `thm:bonding`, `def:sigma`.
- [2] *Cascade-Classical Correspondence IV: Discrete Closure Dynamics*, paper P4. `papers/cascade-closure-dynamics/cascade-closure-dynamics.tex`. Pinned results used here, cited by label: `def:coboundary` (the $F^0 \rightarrow F^1 = \text{Im}(\mathbb{O})$ projection), `def:An` (P4’s full block operator A_n , whose top-left block $d_n^* d_n$ we denote here by the local short-hand A_n^{vertex}), `def:Fn` (P4’s closure functional), and `thm:flow-exists` (style-only: same finite-dimensional ODE / matrix-exponential argument used in Proposition 7.1(O0.3) for L_N^{block}).
- [3] *The Cascade Curvature Operator under Refinement: Schur-Complement Identity and Energy Intertwining*. `papers/cascade-refinement-compat/cascade-refinement-compat.tex`. Pinned results used here, cited by label: `def:abstract-model` (the abstract scalar pure-midpoint refinement model), `def:Atilde` (the scaled curvature operator $\tilde{A}_n = 2^n A_n^{\text{vertex}}$ in that abstract model), `prop:strict-fails` (($O2$) as stated fails for nontrivial refinements), `thm:O3-discharge` (($O3$) closed in the abstract scalar model), `sec:scope-and-gap` (the open lift hypotheses ($L1$) and ($L3$)).
- [4] *Cascade Closure: A Conditional Compatibility Template for Geometric State Selection*, paper P-mech. `papers/cascade-mechanism/cascade-mechanism.tex`. Pinned results used here: the consumer API ($O0$)–($O3$) (§3.4 lines 594–627), with ($O1$) specifically at lines 609–617, and the explicit “selection is the successor’s task” framing at lines 637–639.
- [5] *Cascade-Classical Correspondence II: The Cascade Algebraic Substrate*, paper P2. `papers/cascade-algebraic-substrate/cascade-algebraic-substrate.tex`. Pinned results used here, cited by label: `def:V600` (V_{600} vertex set); `def:icosian-ring` (the icosian ring $I \subset \mathbb{H}$); `prop:24-in-600` ($V_{24} \subset V_{600}$); `tab:P2-spectrum` (the V_{600} scalar Laplacian spectrum, used only in the $N = 0$ worked numerics of §8). **No P2 character data is load-bearing for the main theorem**; irreducibility of the H_4 -reflection representation is taken from [6].
- [6] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics **29**, Cambridge University Press, 1990. H_4 as a finite irreducible Coxeter group: Ch. 2; irreducibility of the standard reflection representation: Ch. 5.

- [7] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics 92, AMS, 1997. Combinatorial graph Laplacian, kernel structure, and operator-norm bound on (multi)graphs: Ch. 1.
- [8] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed., Academic Press, 1980. Spectral theorem and functional calculus for self-adjoint finite-dimensional operators: Ch. VIII.5; bounded operators on Hilbert spaces: Ch. II.
- [9] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013. Matrix exponentials and linear ODEs: Ch. 6.2; spectrum and operator norm of block-diagonal matrices: Ch. 6.4.
- [10] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973. 600-cell edge graph: §22.7; 24-cell: §22.2; Schläfli refinement: §8.